Wave-Tank Calibration and Validation of an Accelerometer-Based Drifting Wave Buoy

Height response, period response, and operating-envelope characterization of the Sargassum drifter wave-sensing module

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Abstract

A low-cost drifting wave buoy — a sealed sensor bottle built around a 6-axis MEMS IMU sampling vertical acceleration at 10 Hz, with on-board spectral processing on an ESP32-S3 — was calibrated and validated in a 10 m laboratory wave flume under monochromatic paddle-generated waves. Ground-truth wave heights were obtained by frame-by-frame video analysis of a wall-mounted ruler at the buoy's station. Across six wave heights ($H = 10$ mm to 48 mm at 0.5 Hz), the on-board significant-wave-height estimate followed the affine relation $H_s = 1.689\,H - 8.48$ mm ($R^2 = 0.9995$, RMSE 0.45 mm), and the inverse map transferred to an independent 0.4 Hz validation point within the ground-truth measurement band. The peak-period estimate was exact to within instrument resolution at all five achievable tank frequencies (0.2 Hz to 1.5 Hz; periods 0.7 s to 5.0 s), with peak-frequency errors $\leq 2\%$. Secondary findings include direct spectral identification of the flume's fundamental seiche at 0.097 Hz (Merian prediction 0.099 Hz at 0.4 m depth), quantification of harmonic content under large paddle strokes (20% of in-band acceleration energy at the first harmonic), an indication of buoy heave resonance near 1 Hz, and practical limits of the facility above 1.5 Hz. The results support field deployment of the module for swell-band period reporting, with height reporting subject to the fitted affine calibration and a per-frequency response correction to be refined in future mixed-sea (wind) testing.

1 Introduction and objectives

The Sargassum drifter is a low-cost expendable surface buoy whose wave-sensing module reports bulk wave parameters — significant wave height (H_s) and peak period (T_p) — over a wireless telemetry link. Before ocean deployment, the module required (i) an absolute height calibration against independently measured waves, (ii) validation of the period estimate across the frequency band of interest, and (iii) characterization of the operating envelope (detection floor, filter-band placement, mooring/tether effects, platform response).

The specific objectives of this test campaign were:

1. Determine the transfer between true crest-to-trough wave height H at the buoy and the on-board estimate H_s, at fixed frequency.

2. Validate the on-board T_p estimate against the wavemaker drive frequency across the achievable range.

3. Verify the height calibration at an independent frequency and amplitude (transfer check).

4. Identify facility and platform artifacts (basin seiche, wave harmonics, tether loading, drift) that bear on data interpretation.

2 Instrument

The buoy is a sealed cylindrical bottle (nominal displacement ∼1 L class) containing:

- **Sensing.** A QMI8658 6-axis MEMS IMU (accelerometer range $\pm 4g$, internal ODR 1 kHz). Vertical (heave) acceleration is derived from attitude-corrected specific force with gravity removed, decimated to 10 Hz.
- **On-board processing.** A sliding window of $N = 1024$ heave samples (102.4 s at 10 Hz) is transformed (FFT); a configurable band-pass $[f_{\mathrm{lo}}, f_{\mathrm{hi}}]$ selects the wave band. Significant wave height is computed from the band-limited variance of the surface-elevation estimate ($H_s = 4\sqrt{m_0}$ convention; Holthuijsen 2007), and T_p from the most prominent spectral peak, gated by a configurable prominence threshold so that no period is reported when no single peak dominates. Estimates update continuously as the window slides.
- **Telemetry.** 2 Hz JSON frames over a persistent WebSocket uplink, each carrying five raw 10 Hz heave samples, the current H_s (mm) and T_p (0.1 s resolution), the spectral peak frequency and prominence, window fill, and housekeeping. Raw heave in every frame permits complete off-line reanalysis independent of the on-board estimates.
- **Remote configuration.** Filter band edges, window length, period-picker parameters, and run annotations are adjustable in situ over the same link, enabling closed-loop calibration without recovering the instrument.

Instrument resolution limits relevant here: H_s is reported in integer millimetres; T_p in 0.1 s steps; the raw spectral bin width is $\Delta f = 10/1024 \approx 0.0098$ Hz.

3 Facility and experimental setup

Tests were conducted in a rectangular glass-walled wave flume: length ≈ 10 m, width 1.22 m (4 ft), still-water depth ≈ 0.4 m (inferred; see §5.4), adjustable and deliberately lowered for the final maximum-wave test. Waves are generated by a voltage-controlled paddle (drive range 0 V to 5 V); a passive absorbing beach terminates the far end. A fixed probe occupies the wavemaker end of the flume; buoy deployments were therefore made at mid-tank.

Ground truth. Wave height at the buoy's station was measured by recording video of a wall-mounted ruler for several wave cycles, then reading crest and trough elevations frame-by-frame; the difference gives the local crest-to-trough height H. For monochromatic waves this equals the wave height without statistical correction. Estimated uncertainty ± 1 mm to 2 mm (parallax, meniscus, frame selection). An earlier protocol using live by-eye ruler readings was found to systematically under-read and all data collected under it were discarded and re-measured (§A).

Buoy station-keeping. For the height sweep the buoy was held mid-tank by a deliberately slack cross-tank tether pair, arranged to restrain along-tank drift while leaving vertical motion free. The tether's slack budget was exceeded (line taut) at wave frequencies ≥ 1.5 Hz; affected runs are flagged. Selected later runs were performed untethered (free-drifting); Stokes drift carried the buoy toward the beach at rates that increased steeply with frequency, exceeding the usable window above ∼2 Hz.

Run protocol. Each test point was bracketed by a timestamped *run* annotation (commanded H, T, drive voltage, configuration notes) stored server-side. Because the estimator window is 102.4 s, each condition was held ≥ 3 minutes and scored on the final 60 s of settled data, with the spectral peak frequency required to match the commanded frequency as an in-condition check. Automated detection of frequency changes (spectral-peak tracking) was used to open brackets when the changed wave physically arrived at the buoy, eliminating operator-timing mislabels.

4 Test series

1. **Height sweep** (fixed 0.5 Hz, $T = 2$ s; tethered): drive stepped 1 V to 5 V, six heights $H = 10, 15, 20, 25, 30, 48$ mm measured at the buoy.
2. **Period sweep** (fixed 3 V): commanded 0.5 → 1.0 → 1.5 Hz; attempts at 2.0 Hz and 2.5 Hz were abandoned when the basin response became irregular (§5.5).
3. **Long-period points** (0.2 Hz, $T = 5$ s, untethered): 3 V and 5 V drives.
4. **Maximum-wave transfer check** (0.4 Hz, $T = 2.5$ s, 5 V = paddle maximum, water level lowered, untethered): independent validation of the height calibration at a different frequency, amplitude, and depth regime.

5 Results

5.1 Height response and affine calibration

Table 1 and Fig. 1 summarize the height sweep. A least-squares affine fit over all six points gives

$$H_s \;=\; a\,H + b, \qquad a = 1.689 \pm 0.02, \quad b = -8.48\,\text{mm}, \tag{1}$$

with $R^2 = 0.9995$ and RMSE 0.45 mm; every residual is below 1 mm. The inverse calibration applied in post-processing and on the live dashboard is $H = (H_s + 8.48)/1.689$.

Table 1: Height sweep at 0.5 Hz ($T = 2.0$ s), slack-tethered, band 0.3 Hz to 2.0 Hz (points 1–4) / 0.3 Hz to 3.0 Hz (points 5–6). H: video-ruler at the buoy. H_s: settled on-board estimate (median of final 60 s). Fit: Eq. (1).

Drive (V)	H (mm)	H_s (mm)	Fit (mm)	Resid. (mm)
1.0	10	9	8.4	0.6
1.5	15	17	16.8	0.2
2.0	20	25	25.3	−0.3
2.5	25	33	33.7	−0.7
3.0	30	42	42.2	−0.2
5.0	48	73	72.6	0.4

Interpretation. For a monochromatic wave of height H, the spectral definition $H_s = 4\sqrt{m_0}$ yields $H_s = \sqrt{2}\,H$ (Holthuijsen, 2007; Longuet-Higgins, 1952). The fitted slope of 1.689 implies the buoy transduced vertical motion ≈ 1.19× the true surface motion at 0.5 Hz — a modest platform-response (RAO) excess consistent with a light, small buoy below but approaching its heave-response peak. The negative intercept corresponds to ≈ 5 mm of surface height that produces no measured response, plausibly a combination of tether restraint, surface tension on the hull, and processing losses at the smallest amplitudes; it implies degraded relative accuracy for waves below ∼10 mm.

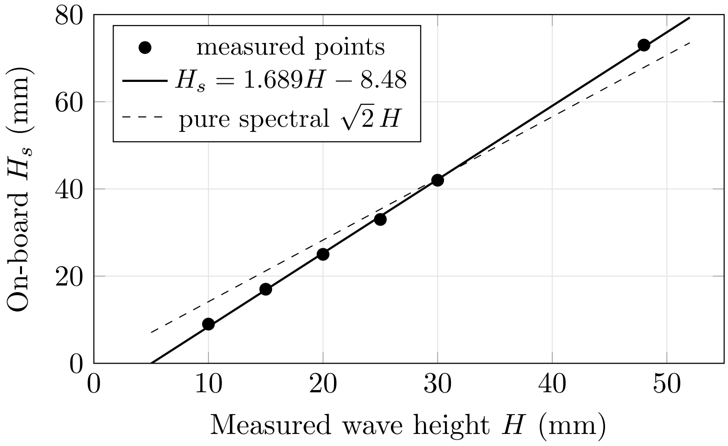

Figure 1: Height response at 0.5 Hz. The fitted slope (1.689) exceeds the monochromatic spectral expectation ($H_s = \sqrt{2}H \approx 1.414H$) by a factor 1.19, attributed to platform response at this frequency; the negative intercept represents a fixed ~5 mm of surface motion (in H units, $-b/a$) not transduced by the buoy.

5.2 Period response

Table 2 lists all frequencies achieved. The on-board period was exact at every point to within the 0.1 s reporting grain, and the underlying spectral peak matched the drive frequency within 2% throughout — including through a taut tether at 1.5 Hz and in the basin-mode regime at 0.2 Hz.

Table 2: Period validation (all at 3 V except as noted). Peak frequency is the median settled on-board spectral peak; T_p the reported period. The 0.667 s case is grain-limited (reported grid is 0.1 s); its spectral peak (1.53 Hz) is within 2%.

Drive freq. (Hz)	T true (s)	T_p (s)	Peak (Hz)	Notes
0.2	5.0	4.9	0.202	untethered; basin-mode regime
0.4	2.5	2.5	0.410	5 V, lowered water
0.5	2.0	2.0	0.510	
1.0	1.0	1.0	1.020	
1.5	0.667	0.7	1.530	tether taut (flagged)

5.3 Transfer validation at 0.4 Hz

With the water level lowered and the paddle at its 5 V maximum, a $T = 2.5$ s wave was generated and measured at the buoy at $H = 30$ mm to 35 mm (video-ruler band). The settled on-board estimate was $H_s = 50$ mm; applying the inverse of Eq. (1) gives

$$\hat{H} = \frac{50 + 8.48}{1.689} = 34.6\,\text{mm},$$

inside the measured band. The calibration fitted at 0.5 Hz therefore transfers to 0.4 Hz at ~1.5× the fitted amplitude mid-range and in a different depth regime, within ground-truth uncertainty.

5.4 Spectral observations

Basin seiche. During early testing with a wide filter band, the H_s estimate was intermittently dominated by energy at 0.097 Hz. Merian's formula for the fundamental longitudinal seiche of a closed rectangular basin, $T = 2L/\sqrt{gd}$ (Rabinovich, 2009), gives 0.099 Hz for $L = 10$ m,

$d = 0.4\,\mathrm{m}$ — identifying the contamination as the tank's fundamental seiche and, incidentally, providing an independent estimate of the working depth. The filter low edge was subsequently placed above the seiche (0.15 Hz to 0.3 Hz depending on test), which removed the contamination. The second longitudinal mode is predicted near 0.2 Hz; long-period forcing at 0.2 Hz to 0.2 Hz consequently produced a partly standing basin response with strong spatial amplitude variation, and long-period height measurements should be regarded as position-specific.

Harmonic content at large stroke. At 3 V drive and 0.5 Hz, off-line spectral decomposition of the raw 10 Hz heave record showed 70.8% of in-band acceleration energy at the fundamental (0.51 Hz), 20.1% at the first harmonic (1.0 Hz), and 6.1% above — the signature of finite-amplitude (Stokes-type) wave steepening rather than facility noise (cross-tank sloshing at the predicted 0.8 Hz band contributed only 1.3%). Because the on-board height estimate converts acceleration to elevation at a single reference frequency, harmonic energy is over-weighted at conversion ($\propto (f/f_p)^4$); a per-bin conversion is recommended (§8).

Platform resonance indication. At fixed 3 V drive, the sensed H_s at 1.0 Hz (83 mm) was double the 0.5 Hz value (42 mm). Without an independent height measurement at 1.0 Hz the wavemaker transfer and platform response cannot be separated, but combined with morning free-float observations (buoy heave $\approx 1.4\times$ surface at $T = 1\,\mathrm{s}$), a buoy heave resonance in the vicinity of 1 Hz is the working hypothesis. A response-amplitude-operator (RAO) determination with independent wave-probe ground truth versus frequency is the definitive follow-up (Raghukumar et al., 2019).

5.5 Operating-envelope findings

- **Facility ceiling.** Above ∼1.5 Hz the basin response became irregular (short waves reflect strongly from an absorbing beach tuned for longer waves); 2.0 Hz forcing produced no stable spectral peak at the buoy and the points were abandoned. This is a facility limit, not an instrument limit.

- **Tether loading.** The slack cross-tank tether went taut at ≥1.5 Hz (drift force grows steeply with frequency). The period estimate was unaffected; height data under taut tether are flagged damped/suspect.

- **Free-drift envelope.** Untethered, the buoy crossed the available half-tank in under 3 min at 2 Hz (faster than one estimator window), but drifted slowly enough at ≤0.5 Hz for clean untethered points.

- **Detection floor.** The smallest wave tested ($H = 10\,\mathrm{mm}$, peak acceleration $\sim 0.05\,\mathrm{m\,s^{-2}}$) was detected with strong spectral prominence (period lock throughout); the practical floor is set by the affine intercept rather than by detection.

- **Run hygiene.** Automated frequency-detection bracketing eliminated two classes of labeling error observed under manual timing (operator-missed steps; clock skew). An instrument-side window reset on run start was identified as unnecessary dead time and disabled server-side mid-campaign; transitions are instead excluded in scoring.

6 Discussion

The instrument's period channel is field-ready: five frequencies spanning 0.2 Hz to 1.5 Hz (periods 0.7 s to 5 s) were reported exactly at instrument resolution, through tether loading, harmonic contamination, and basin-mode conditions. This is the primary quantity for the sargassum-drift application.

The height channel is usable with the affine calibration of Eq. (1), which held with sub-millimetre residuals over a 5× height range and transferred to an independent frequency/amplitude/depth point. Two caveats bound its validity: (i) the slope contains a platform-response factor measured at 0.4 Hz to 0.5 Hz; near the suspected ∼1 Hz resonance the effective slope is larger (under-estimating true height if the tank calibration is applied unmodified), and at long ocean swell periods (≥5 s) the platform follows the surface nearly perfectly, so the slope should relax toward the spectral $\sqrt{2}$; a per-frequency correction table is the appropriate refinement. (ii) the negative intercept implies a ∼5 mm height floor below which relative error grows rapidly — immaterial for ocean swell but relevant to tank work.

All conclusions rest on monochromatic seas. The estimator's field task — extracting the energetic peak from a mixed swell-plus-wind-sea spectrum — was not exercised here and is the highest-value follow-up test (the facility can superimpose wind chop on paddle swell).

7 Conclusions

1. Peak-period reporting is validated across 0.7 s to 5.0 s with $\leq 2\%$ spectral-peak error at every achievable facility frequency.
2. On-board H_s follows an affine law in true height at 0.5 Hz: $H_s = 1.689H - 8.48$ mm ($R^2 = 0.9995$); the inverse map recovers true height to ∼1 mm in-sample and validated at 0.4 Hz within ground-truth uncertainty.
3. The flume's fundamental seiche (0.097 Hz measured; 0.099 Hz Merian) was identified and excluded by filter-band placement; band edges must bracket the wave band with margin at both ends (two edge-collision incidents during the campaign).
4. Facility limits: clean monochromatic operation 0.2 Hz to 1.5 Hz; irregular above; long-period forcing is basin-mode dominated and position-dependent.
5. Platform heave response likely peaks near 1 Hz; an RAO determination and mixed-sea (wind) validation are the recommended next campaigns.

8 Recommendations

1. **Firmware: runtime calibration constants.** Expose the affine pair (a, b) as persisted, remotely settable parameters applied at reporting time (raw H_s retained in telemetry), replacing the current display-side application.
2. **Firmware: per-bin spectral conversion.** Convert acceleration to elevation per FFT bin ($/\omega^4$ in power) before band integration, removing the harmonic over-weighting of §5.4.
3. **Firmware: remove run-start window reset** (already neutralized operationally) and add band-edge sanity warnings when the spectral peak approaches either filter edge.
4. **Testing: RAO campaign** — independent wave-probe ground truth versus frequency at fixed height, to convert the single-point platform-response factor into a response curve.
5. **Testing: mixed-sea validation** — paddle swell plus wind chop; verify the prominence-gated T_p locks the energetic swell peak and characterize H_s behaviour in bimodal spectra.
6. **Ocean preset.** For deployment, shift the filter band to the ocean swell/wind-sea range (∼0.05 Hz to 0.5 Hz); re-derive the platform-response slope for long-period swell (expected $\to \sqrt{2}$).

References

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A Campaign notes and discarded data

An initial height sweep (six points, 10 mm to 48 mm nominal) was conducted with live by-eye ruler readings as ground truth. Midway through the campaign the field team determined that this method systematically under-read wave height; the entire first sweep was voided and repeated with the video-ruler protocol. Data from the voided sweep were internally self-consistent (the on-board estimates tracked a straight line) but are excluded from all fits. One period-sweep point was additionally re-measured after an operator-side missed step was caught by the instrument itself (the telemetry showed the spectral peak unchanged at the prior frequency), motivating the switch to detection-based run bracketing used for the rest of the campaign.

B Board configuration

Sampling 10 Hz; window $N = 1024$ (102.4 s); band-pass low edge 0.08 / 0.15 / 0.30 Hz and high edge 0.5 / 2.0 / 3.0 Hz across campaign phases (final long-period configuration 0.15 Hz to 3.0 Hz); T_p prominence gate 10.0 (default), median filter $n = 5$; telemetry 2 Hz; timestamps server-side at receipt.